COMSATS University Islamabad
Department of Computer Science

Course Information	
Course Code: **CSC354**	Course Title: **Machine Learning**
Credit Hours: **3(3,0)**	Lecture Hours/Week: **3**
Lab Hours/Week: **0**	Pre-Requisites: **None**

Text and Reference Books

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

Reference Books:

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, Aurelien Geron, O`Reilly Media, 2017.
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.

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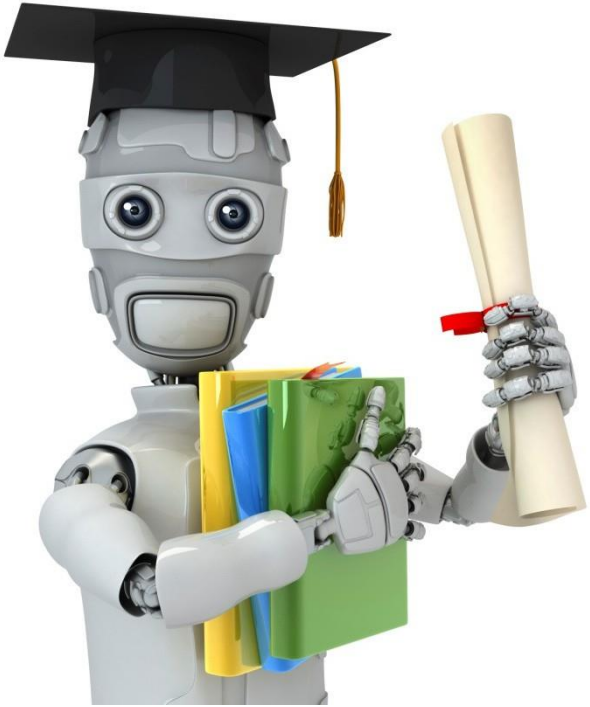
- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on (zeenatzulfiqar@cuiatd.edu.pk)



Machine Learning

Supervise Learning:

KNN Regression

Question (KNN Regression with 3 Features)

dataset contains:

- **X_1 = Hours Studied**
- **X_2 = Sleep Hours**
- **X_3 = Attendance (%)**
- **Y = Exam Score**

X_1	X_2	X_3	Y
2	6	60	50
3	5	65	55
5	7	70	65
6	8	80	75
4	6	75	60

Predict score for:

(X_1=5, X_2=6, X_3=72) using **k = 3**

Step 1: Distance Formula

We use Euclidean distance:

$$d = \sqrt{(x_1 - X_1)^2 + (x_2 - X_2)^2 + (x_3 - X_3)^2}$$

Step 2: Calculate Distance

From (2,6,60)

$$= \sqrt{(5-2)^2 + (6-6)^2 + (72-60)^2}$$

$$= \sqrt{9 + 0 + 144} = \sqrt{153} \approx 12.37$$

From (3,5,65)

$$= \sqrt{(5-3)^2 + (6-5)^2 + (72-65)^2}$$

$$= \sqrt{4 + 1 + 49} = \sqrt{54} \approx 7.35$$

From (5,7,70)

$$= \sqrt{0 + 1 + 4} = \sqrt{5} \approx 2.23$$

From (6,8,80)

$$= \sqrt{1 + 4 + 64} = \sqrt{69} \approx 8.30$$

From (4,6,75)

$$= \sqrt{1+0+9} = \sqrt{10} \approx 3.16$$

Step 3: Select k = 3 Nearest

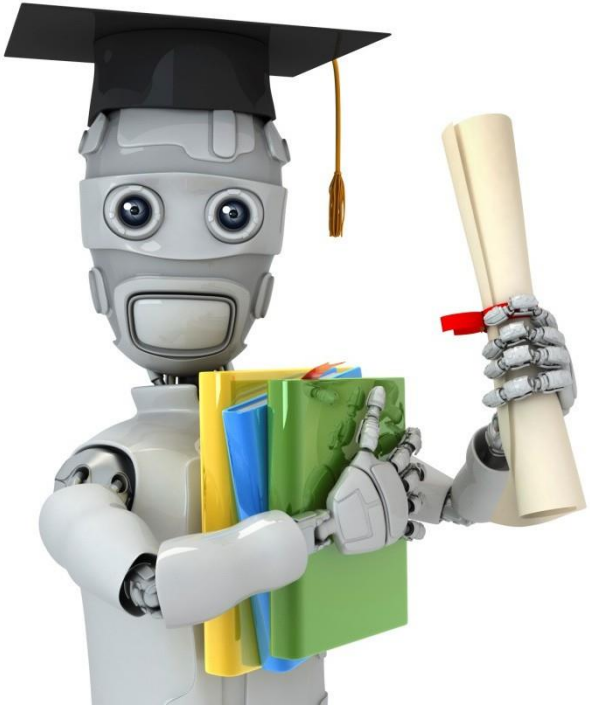
Smallest distances:

- (5,7,70) → Y = 65
- (4,6,75) → Y = 60
- (3,5,65) → Y = 55

Step 4: Compute Average

$$Y_{pred} = \frac{65+60+55}{3} = \frac{180}{3} = 60$$

Predicted Exam Score = 60



Machine Learning

Supervise Learning:

Regression Model

Regression Model

- Regression modeling is a process of determining a relationship between one or more independent variables and one dependent or output variable.
- **Examples:**
  1. Predicting the height of a person given the age of the person.
  2. Predicting the price of the car given the car model, year of manufacturing, mileage, engine capacity, etc.

Regression Model

- Based on the type of functions used to represent the relationship between the dependent or output variable and independent variables, the regression models are categorized into four types. The regression models are,

1. Simple Linear Regression
2. Multiple Regression
3. Polynomial Regression
4. Logistic Regression

Regression Types

- **Simple Linear Regression** is a supervised machine learning technique used to model and analyze the relationship between one or more independent variables and a continuous dependent variable by fitting a linear equation to observed data.
- **Simple Linear Regression** is a method used to find the relationship between:
- **one independent variable (X)** and **one dependent variable (Y)**
- It fits a **straight line** to predict Y from X.

Simple Linear Regression

$y=mx+b$

Where:

- **y** = predicted value
- **x** = input variable
- **m** = slope (rate of change)
- **b** = intercept (starting point)

Simple Linear Regression Working

Step 1: Collect Data
•Gather pairs of values (X, Y)
•Example: Hours studied & exam score
Step 2: Plot the Data
•Draw points on a graph
•X → horizontal axis
•Y → vertical axis

Step 3: Find Best-Fit Line
•Draw a straight line that best fits all points
•This line minimizes error (difference between actual & predicted values)
Step 4: Calculate Slope (m)
•Shows how much Y changes when X increases
Step 5: Calculate Intercept (b)
•Value of Y when X = 0
Step 6: Form Equation
•Put values of m and b in equation
•Get final regression model
Step 7: Make Prediction
•Put new X value in equation
•Get predicted Y

Question#1: Simple Linear Regression

Problem Statement:
A teacher records the **number of hours students studied** and their **exam scores**. Predict the exam score of a student based on hours studied using **Linear Regression**.

Hours Studied (X)	Exam Score (Y)
1	2
2	3
4	3
3	4
5	5

Simple Linear Regression

Step 1: Prepare the table for calculations

We need: X, Y, X^2 ,XY

X	Y	X^2	XY
1	2	1	2
2	3	4	6
4	3	16	12
3	4	9	12
5	5	25	25

Simple Linear Regression

Now calculate the sums:

$$\sum X = 1 + 2 + 4 + 3 + 5 = 15$$

$$\sum Y = 2 + 3 + 3 + 4 + 5 = 17$$

$$\sum X^2 = 1 + 4 + 16 + 9 + 25 = 55$$

$$\sum XY = 2 + 6 + 12 + 12 + 25 = 57$$

Number of points: $N = 5$

Simple Linear Regression

Step 2: Calculate slope m

$$m = \frac{N\sum XY - (\sum X)(\sum Y)}{N\sum X^2 - (\sum X)^2}$$

Substitute values:

$$m = \frac{5(57) - (15)(17)}{5(55) - (15)^2} = \frac{285 - 255}{275 - 225} = \frac{30}{50} = 0.6$$

Simple Linear Regression

Step 3: Calculate intercept b

$$b = \frac{\sum Y - m\sum X}{N} = \frac{17 - (0.6 \cdot 15)}{5} = \frac{17 - 9}{5} = \frac{8}{5} = 1.6$$

Step 4: Write the regression equation

$$Y = 0.6X + 1.6$$

Example: If a student studies **6 hours**:

$$Y = 0.6(6) + 1.6 = 5.2$$

linear regression

Question#2: Find linear regression equation for the following two sets of data:

x	2	4	6	8
y	3	7	5	10

Solution Simple linear regression

x	y	x^2	xy
2	3	4	6
4	7	16	28
6	5	36	30
8	10	64	80
$\sum x$ = 20	$\sum y$ = 25	$\sum x^2$ = 120	$\sum xy$ = 144

Solution Simple linear regression

$$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

$$b = \frac{4\times 144 - 20\times 25}{4\times 120 - 400}$$

b = 0.95

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$a = \frac{25\times 120 - 20\times 144}{4(120) - 400}$$

a = 1.5

Linear regression is given by:

y = a + bx

y = 1.5 + 0.95 x

Question#3

X	Y
1	1
2	2
3	1.3
4	3.75
5	2.25

Calculation Table

X	Y	X^2	XY
1	1	1	1
2	2	4	4
3	1.3	9	3.9
4	3.75	16	15
5	2.25	25	11.25

Simple Linear Regression

Find Sums

- $\Sigma X = 15$
- $\Sigma Y = 10.3$
- $\Sigma X^2 = 55$
- $\Sigma XY = 35.15$
- $n = 5$

Calculate Slope (m)

$$m = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{n\Sigma X^2 - (\Sigma X)^2}$$

$$m = \frac{5(35.15) - (15)(10.3)}{5(55) - (15)^2}$$

$$m = \frac{175.75 - 154.5}{275 - 225} = \frac{21.25}{50} = 0.425$$

Simple Linear Regression

Calculate Intercept (b)

$$b = \frac{\Sigma Y - m\Sigma X}{n}$$

$$b = \frac{10.3 - (0.425 \times 15)}{5}$$

$$b = \frac{10.3 - 6.375}{5} = \frac{3.925}{5} = 0.785$$

Final Regression Equation

$$Y = 0.425X + 0.785$$

Simple Linear Regression

Example Prediction

For **X = 6**

$$Y = 0.425(6) + 0.785$$

$$= 2.55 + 0.785 = 3.335$$

Error Calculation in Simple Linear Regression

In regression, **error** means:
Difference between **actual value (Y)** and **predicted value (Ŷ)**

Error (Residual)

$$\text{Error} = Y - \hat{Y}$$

- **Y** = actual value
- **Ŷ** = predicted value

Using Model
previous solution:

$$\hat{Y} = 0.425X + 0.785$$

Calculate Predicted Values ($\hat{Y}$)

X	Y (Actual)	$\hat{Y}$ = 0.425X+0.785	Error (Y - $\hat{Y}$)
1	1	1.21	-0.21
2	2	1.635	0.365
3	1.3	2.06	-0.76
4	3.75	2.485	1.265
5	2.25	2.91	-0.66

Types of Error Measures

(a) Mean Absolute Error (MAE)

$$MAE = \frac{\sum |Y - \hat{Y}|}{n}$$

$$= \frac{0.21 + 0.365 + 0.76 + 1.265 + 0.66}{5} = \frac{3.26}{5} = 0.652$$

(b) Mean Squared Error (MSE)

$$MSE = \frac{\sum (Y - \hat{Y})^2}{n}$$

$$= \frac{0.044 + 0.133 + 0.578 + 1.600 + 0.436}{5} = \frac{2.791}{5} \approx 0.558$$

(c) Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE} = \sqrt{0.558} \approx 0.747$$

Types of Error Measures

- Error = difference between real & predicted
- MAE → average error
- MSE → squared error (penalizes big mistakes)
- RMSE → square root (easy to understand scale)

Multiple Linear Regression(MLR)

- **Multiple Linear Regression extends simple linear regression** by using **two or more independent variables (X_1, X_2, …)** to predict a **continuous dependent variable.** Its equation is $\boldsymbol{Y = b_0 + b_1 X_1 + b_2 X_2 + \cdots + \epsilon}$
- This allows the model to capture more complex relationships between predictors and the target.
- It is used in applications like **predicting house prices based on size, bedrooms, and location.** The model assumes linearity, no multicollinearity, and normally distributed residuals.

Multiple Linear Regression(MLR)

Size (X_1)	Bedrooms (X_2)	Price (Y)
1	3	2
2	4	5
3	7	9

Multiple Linear Regression(MLR)

Step 1: Model Equation

$$Y = b_0 + b_1X_1 + b_2X_2$$

Step 2: Form Equations

From each row:

$2 = b_0 + 1b_1 + 3b_2$ ----------→1

$5 = b_0 + 2b_1 + 4b_2$ ----------→2

$9 = b_0 + 3b_1 + 7b_2$ ----------→3

Multiple Linear Regression(MLR)

Step 3: Eliminate b_0

Subtract (1) from (2):

$5 - 2 = (2-1)b_1 + (4-3)b_2$

$3 = b_1 + b_2$ … (A)

Subtract (2) from (3):

$9 - 5 = (3-2)b_1 + (7-4)b_2$

$4 = b_1 + 3b_2$ … (B)

Multiple Linear Regression(MLR)

Step 4: Solve for b_1 and b_2

From (A):

$b_1 = 3 - b_2$

Put into (B):

$4 = (3 - b_2) + 3b_2$

$4 = 3 + 2b_2$

$2b_2 = 1$

$b_2 = 0.5$

Now find b_1:

$b_1 = 3 - 0.5 = 2.5$

Multiple Linear Regression(MLR)

Step 5: Find b_0

Put values into equation (1):

$2 = b_0 + 1(2.5) + 3(0.5)$

$2 = b_0 + 2.5 + 1.5$

$2 = b_0 + 4$

$b_0 = -2$

Final Model

$$Y = -2 + 2.5X_1 + 0.5X_2$$

Multiple Linear Regression(MLR)

Step 6: Check (Verification)

Take row 3:

Y = -2 + 2.5(3) + 0.5(7)

Y = -2 + 7.5 + 3.5 = **9**

Final Answer

- b_0 = **-2**
- b_1 = **2.5**
- b_2 = **0.5**

Method#2:Multiple Linear Regression(MLR)

Step 1: Given Dataset

X_1	X_2	Y
1	3	2
2	4	5
3	7	9

Method#2:Multiple Linear Regression(MLR)

Step 2: Matrix Formula

$$\beta = (X^T X)^{-1} X^T Y$$

Step 3: Construct Matrices

X matrix (add column of 1 for intercept b_0)

$$X = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & 7 \end{bmatrix}$$

Y matrix

$$Y = \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}$$

Method#2:Multiple Linear Regression(MLR)

Step 4: Find X^T

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 4 & 7 \end{bmatrix}$$

Step 5: Compute $X^T X$

$$= \begin{bmatrix} (1\cdot 1 + 1\cdot 1 + 1\cdot 1) & (1\cdot 1 + 1\cdot 2 + 1\cdot 3) & (1\cdot 3 + 1\cdot 4 + 1\cdot 7) \\ (1\cdot 1 + 2\cdot 1 + 3\cdot 1) & (1\cdot 1 + 2\cdot 2 + 3\cdot 3) & (1\cdot 3 + 2\cdot 4 + 3\cdot 7) \\ (3\cdot 1 + 4\cdot 1 + 7\cdot 1) & (3\cdot 1 + 4\cdot 2 + 7\cdot 3) & (3\cdot 3 + 4\cdot 4 + 7\cdot 7) \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 3 & 6 & 14 \\ 6 & 14 & 32 \\ 14 & 32 & 74 \end{bmatrix}$$

Method#2:Multiple Linear Regression(MLR)

Step 6: Compute $X^T Y$
(Row × Column)

$$= \begin{bmatrix} (1 \cdot 2 + 1 \cdot 5 + 1 \cdot 9) \\ (1 \cdot 2 + 2 \cdot 5 + 3 \cdot 9) \\ (3 \cdot 2 + 4 \cdot 5 + 7 \cdot 9) \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 16 \\ 39 \\ 89 \end{bmatrix}$$

Method#2:Multiple Linear Regression(MLR)

Step 7: Find $(X^T X)^{-1}$

$$A = X^T X = \begin{bmatrix} 3 & 6 & 14 \\ 6 & 14 & 32 \\ 14 & 32 & 74 \end{bmatrix}$$

Find **Cofactor Matrix**

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Method#2:Multiple Linear Regression(MLR)

Determinant

$$|A| = 3\begin{vmatrix} 14 & 32 \\ 32 & 74 \end{vmatrix} - 6\begin{vmatrix} 6 & 32 \\ 14 & 74 \end{vmatrix} + 14\begin{vmatrix} 6 & 14 \\ 14 & 32 \end{vmatrix}$$

$$= 3(1036 - 1024) - 6(444 - 448) + 14(192 - 196)$$

$$= 3(12) - 6(-4) + 14(-4) = 36 + 24 - 56 = 4$$

Method#2:Multiple Linear Regression(MLR)

Step 1: Write in Cofactor Form

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

Method#2:Multiple Linear Regression(MLR)

Step 2: Compute Each Cofactor (Matrix Form)

First Row

$$C_{11} = + \begin{vmatrix} 14 & 32 \\ 32 & 74 \end{vmatrix} = 14 \cdot 74 - 32 \cdot 32 = 12$$

$$C_{12} = - \begin{vmatrix} 6 & 32 \\ 14 & 74 \end{vmatrix} = -(444 - 448) = 4$$

$$C_{13} = + \begin{vmatrix} 6 & 14 \\ 14 & 32 \end{vmatrix} = 192 - 196 = -4$$

Method#2:Multiple Linear Regression(MLR)

Second Row

$$C_{21} = -\begin{vmatrix} 6 & 32 \\ 14 & 74 \end{vmatrix} = 4$$

$$C_{22} = +\begin{vmatrix} 3 & 14 \\ 14 & 74 \end{vmatrix} = 222 - 196 = 26$$

$$C_{23} = -\begin{vmatrix} 3 & 6 \\ 14 & 32 \end{vmatrix} = -(96 - 84) = -12$$

Method#2:Multiple Linear Regression(MLR)

Third Row

$$C_{31} = + \begin{vmatrix} 6 & 14 \\ 14 & 32 \end{vmatrix} = -4$$

$$C_{32} = - \begin{vmatrix} 3 & 14 \\ 6 & 32 \end{vmatrix} = -(96 - 84) = -12$$

$$C_{33} = + \begin{vmatrix} 3 & 6 \\ 6 & 14 \end{vmatrix} = 42 - 36 = 6$$

Method#2:Multiple Linear Regression(MLR)

Final Cofactor Matrix

$$C = \begin{bmatrix} 12 & 4 & -4 \\ 4 & 26 & -12 \\ -4 & -12 & 6 \end{bmatrix}$$

Method#2:Multiple Linear Regression(MLR)

Signs follow pattern:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Method#2:Multiple Linear Regression(MLR)

Inverse

$$A^{-1} = \frac{1}{4}C$$

$$= \begin{bmatrix} 3 & 1 & -1 \\ 1 & 6.5 & -3 \\ -1 & -3 & 1.5 \end{bmatrix}$$

Method#2:Multiple Linear Regression(MLR)

Step 8: Final Multiplication

$$\beta = (X^T X)^{-1} X^T Y$$

$$\beta = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 6.5 & -3 \\ -1 & -3 & 1.5 \end{bmatrix} \begin{bmatrix} 16 \\ 39 \\ 89 \end{bmatrix} = \begin{bmatrix} (3 \cdot 16 + 1 \cdot 39 + (-1) \cdot 89) \\ (1 \cdot 16 + 6.5 \cdot 39 + (-3) \cdot 89) \\ ((-1) \cdot 16 + (-3) \cdot 39 + 1.5 \cdot 89) \end{bmatrix}$$

$$\beta = \begin{bmatrix} -2 \\ 2.5 \\ 0.5 \end{bmatrix}$$

Method#2:Multiple Linear Regression(MLR)

Final Interpretation

Regression equation becomes:

$$y = -2 + 2.5x_1 + 0.5x_2$$

Method#2:Multiple Linear Regression(MLR)

Final Answer

- $b_0 = -2$
- $b_1 = 2.5$
- $b_2 = 0.5$

Always remember steps:

1. Make **X matrix (with 1 column)**
2. Compute $\mathbf{X^T}$
3. Find $\mathbf{X^TX}$
4. Take **inverse**
5. Multiply with $\mathbf{X^TY}$

Logistic Regression

What is Logistic Regression?

Logistic Regression is a **supervised machine learning algorithm** used for **classification problems**.

It predicts the **probability** that an input belongs to a class.

✓ Output is always between **0 and 1**

✓ Used mainly for **binary classification**

Example:

•0 → Fail, 1 → Pass

•0 → No disease, 1 → Disease

Logistic Regression

Key Idea
Instead of predicting a number like **linear regression,** it predicts a **probability (0 to 1)**.

- Take input features (X_1, X_2, …)
- Multiply with weights (b_1, b_2, …)
- Add bias (b_0)
- Pass result into sigmoid function
- Get probability (0 to 1)
- Apply threshold:
- ≥ 0.5 → Class 1
- < 0.5 → Class 0

Step 1: Create Dataset

We classify **Pass (1) / Fail (0)** based on **Study Hours (X)**

X (Hours)	Y (Result)
1	0
2	0
3	0
4	1
5	1

Logistic Regression

Step 2: Model

$$z = b_0 + b_1 X$$

Sigmoid:

$$P = \frac{1}{1 + e^{-z}}$$

Step 3: Initialize

Let:

- $b_0 = 0$
- $b_1 = 0$
- Learning rate $\alpha = 0.1$

Logistic Regression

Step 4: Iteration 1

Since weights = 0:

$$z = 0, P = 0.5$$

Logistic Regression

X	Y	P	Error (Y−P)
1	0	0.5	-0.5
2	0	0.5	-0.5
3	0	0.5	-0.5
4	1	0.5	+0.5
5	1	0.5	+0.5

Logistic Regression

Step 5: Update Weights
Formula:

$$b_1 = b_1 + \alpha \sum (Error \cdot X)$$

$$b_0 = b_0 + \alpha \sum (Error)$$

Calculate
Σ(Error):

$$-0.5 - 0.5 - 0.5 + 0.5 + 0.5 = -0.5$$

Logistic Regression

Σ(Error × X):

$$(-0.5 \times 1) + (-0.5 \times 2) + (-0.5 \times 3) + (0.5 \times 4) + (0.5 \times 5)$$
$$= (-0.5 - 1 - 1.5) + (2 + 2.5)$$
$$= -3 + 4.5 = 1.5$$

Update:

$$b_0 = 0 + 0.1(-0.5) = -0.05$$
$$b_1 = 0 + 0.1(1.5) = 0.15$$

Step 6: Iteration 2

Now:

- $b_0 = -0.05$
- $b_1 = 0.15$

Logistic Regression

Compute z and P

Example:

For X = 1:

$$z = -0.05 + 0.15(1) = 0.10 \Rightarrow P \approx 0.525$$

For X = 5:

$$z = -0.05 + 0.15(5) = 0.70 \Rightarrow P \approx 0.67$$

Observation

•Small X → probability near 0.5

•Large X → probability increases

Model is **learning pattern**

Logistic Regression

Step 7: (Concept of More Iterations)

Repeat:

- Compute P
- Find error
- Update weights

After many iterations:

- $X \leq 3 \rightarrow$ Class 0
- $X \geq 4 \rightarrow$ Class 1

Logistic Regression

Tips

Write:

1. Dataset
2. Model (z + sigmoid)
3. Initial weights
4. One iteration (fully solved)
5. Update formula
6. Conclusion

polynomial Regression

polynomial Regression is a type of regression where:

- Relationship between X and Y is **non-linear**
- But we still use a **linear model with transformed features**

Instead of straight line, we fit a **curve**

polynomial Regression

2. Why Do We Need It?

Linear Regression fails when data is like:

- Curved (U-shape, parabola)
- Non-linear relationship

Example:

- Growth rate
- Temperature vs time
- Price vs demand

polynomial Regression

3. When to Use?

✓ When data shows **curved pattern**

✓ Linear regression cannot fit properly

Example:

- Growth rate
- Temperature changes
- Stock trends

polynomial Regression

3. Polynomial Equation
Degree 2 (Quadratic):

$$Y = b_0 + b_1 X + b_2 X^2$$

Degree 3:

$$Y = b_0 + b_1 X + b_2 X^2 + b_3 X^3$$

4. Example Dataset

X	Y
1	1
2	4
3	9
4	16

Pattern:

$$Y = X^2$$

Polynomial Regression using Matrix Method

We use:

$$\hat{A} = (X^T X)^{-1} X^T Y$$

x	y
1	2
2	5
3	10
4	17
5	26

We choose **degree = 2 polynomial regression**

Polynomial Regression using Matrix Method

Step 2: Model Form

$$y = a_0 + a_1 x + a_2 x^2$$

So we need:

$$A = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

Step 3: Design Matrix (X)

Each row = $[1, x, x^2]$

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \end{bmatrix}$$

Polynomial Regression using Matrix Method

Step 4: Output Vector (Y)

$$Y = \begin{bmatrix} 2 \\ 5 \\ 10 \\ 17 \\ 26 \end{bmatrix}$$

Step 5: Compute $X^T X$

Transpose: $X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 9 & 16 & 25 \end{bmatrix}$

Now multiply:

Each entry:

- Sum(1) = 5
- Sum(x) = 15
- Sum(x^2) = 55
- Sum(x^3) = 225
- Sum(x^4) = 979

So: $X^T X = \begin{bmatrix} 5 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix}$

Polynomial Regression using Matrix Method

Step 6: Compute $X^T Y$
Calculate step by step:
•Σy = 2 + 5 + 10 + 17 + 26 = 60
•Σxy = 2 + 10 + 30 + 68 + 130 = 240
•Σx²y = 2 + 20 + 90 + 272 + 650 = 1034
So:

$$X^T Y = \begin{bmatrix} 60 \\ 240 \\ 1034 \end{bmatrix}$$

Step 7: Solve Normal Equation

$$(X^T X)A = X^T Y$$

So:

$$\begin{bmatrix} 5 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 60 \\ 240 \\ 1034 \end{bmatrix}$$

Polynomial Regression using Matrix Method

STEP 9: Solve Using Inverse Method

$$A = (X^T X)^{-1} X^T Y$$

Let:

$$M = \begin{bmatrix} 5 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix}$$

Polynomial Regression using Matrix Method

Step 9.1: Determinant

$$| M | = 500$$

Step 9.2: Cofactor Matrix

$$\text{Cof}(M) = \begin{bmatrix} 2140 & -410 & 20 \\ -410 & 70 & -10 \\ 20 & -10 & 5 \end{bmatrix}$$

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Step 9.3: Adjoint (Transpose of Cofactor)

$$\text{Adj}(M) = \begin{bmatrix} 2140 & -410 & 20 \\ -410 & 70 & -10 \\ 20 & -10 & 5 \end{bmatrix}$$

Step 9.4: Inverse

$$M^{-1} = \frac{1}{500}\begin{bmatrix} 2140 & -410 & 20 \\ -410 & 70 & -10 \\ 20 & -10 & 5 \end{bmatrix}$$

STEP 10: Multiply $M^{-1}X^TY$

$$A = M^{-1}X^TY$$

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compute a_0

$$a_0 = \frac{1}{500}\big(2140(60) - 410(240) + 20(1034)\big)$$
$$= \frac{1}{500}(128400 - 98400 + 20680)$$
$$= \frac{50000}{500} = 1$$

Compute a_1

$$a_1 = \frac{1}{500}\big(-410(60) + 70(240) - 10(1034)\big)$$
$$= \frac{1}{500}(-24600 + 16800 - 10340)$$
$$= \frac{-18140}{500} = 0$$

Compute a_2

$$a_2 = \frac{1}{500}\big(20(60) - 10(240) + 5(1034)\big)$$
$$= \frac{1}{500}(1200 - 2400 + 5170)$$
$$= \frac{3970}{500} = 1$$

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INAL RESULT

$$A = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

FINAL MODEL

$$\boxed{y = 1 + x^2}$$

1. Simple Linear Regression

•One independent variable (X) and one dependent variable (Y)

•Example: predicting salary from experience

2. Multiple Linear Regression

•More than one independent variable

•Example: predicting house price using size, rooms, location

3. Polynomial Regression

•Relationship between X and Y is non-linear (curved line)

•Uses powers of X (X^2, X^3, etc.)

4. Logistic Regression

•Used for **classification (not continuous values)**

•Output is probability (0 to 1)

•Example: spam or not spam

Why is it called Logistic "Regression" but used for Classification?

1. Historical reason (main reason)
Logistic regression was developed as a **statistical regression method**, not as a classification algorithm.
It was designed to model a relationship between variables, just like linear regression.

2. It actually *does regression first*

Logistic regression first calculates a **linear equation**:

•It computes a value like:

$$z = wX + b$$

This part is exactly like **linear regression (continuous output)**.

3. Then it transforms output using sigmoid function

Instead of giving final number, it applies:

•**Sigmoid function**

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

This converts result into a **probability (0 to 1)**.

4. Then classification happens

•If probability ≥ 0.5 → Class 1

•If probability < 0.5 → Class 0

So final output becomes **classification**, not regression.

- It **uses regression equation internally**
- But **final purpose is classification**

That's why:

It is called "Logistic Regression" (because of regression equation)

But it is used for "Classification" (because output is class)